

L1.4 The Derivative

§2.1–2.2 — what you need to know

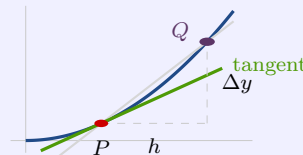
The derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

“take the difference quotient, and let h go to zero”

The fraction on its own is the **difference quotient** — the slope of the **secant** through P and Q . Shrink h and Q slides toward P .

At $h = 0$ there is **no second point** and **no secant at all**. That is why it has to be a *limit* and not an evaluation.



$$f'(x) = \frac{dy}{dx} = \frac{d}{dx}(f(x)) \quad \text{evaluate with } \left. \frac{dy}{dx} \right|_{x=3} \quad \text{second derivative } f''(x) = \frac{d^2y}{dx^2}$$

dx and dy are called **differentials** — $\frac{dy}{dx}$ is not one symbol. The book also writes $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ and $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$; same derivative, and we compute with the boxed one.

EQUATION OF THE TANGENT LINE

The derivative is a slope. If the question asks for the *line*, you must find the tangent line. For $f(x) = 2x^2 + 1$ at $x = 2$: $f'(2) = 8$ and $f(2) = 9$, so

$$y = mx + b : \quad 9 = 8(2) + b \Rightarrow b = -7 \quad \Rightarrow \quad \boxed{y = 8x - 7}$$

Keep **secant slope**, **tangent slope**, and **the equation of the tangent line** separate in your head.

THE GRAPH OF f' FROM THE GRAPH OF f

f' is a graph of the slopes.

If f has a HT $\rightarrow f'$ has a **zero** f increasing $\rightarrow f' > 0$

If f has an IP $\rightarrow f'$ has a HT f decreasing $\rightarrow f' < 0$

f'' is the **concavity** — how strongly the curve is bent. Arcing downward gives $f'' < 0$, **cup down, or CD**; arcing upward gives $f'' > 0$, **cup up, or CU**. *Make the cup with your hands.* Where the concavity switches is an **inflection point**.

POSITION, VELOCITY, ACCELERATION

Position $x(t)$ — where you are, measured from a chosen origin. **Velocity** is its derivative, and **acceleration** is the derivative of that.

$$v(t) = x'(t) = \frac{dx}{dt} \qquad a(t) = v'(t) = x''(t) = \frac{d^2x}{dt^2}$$

For $x(t) = t^3 - t$: $v = 3t^2 - 1$ and $a = 6t$.

$\frac{\Delta x}{\Delta t}$ over an interval is **average** velocity (a secant slope); $x'(t)$ at one instant is **instantaneous** velocity (a tangent slope). Velocity is signed: negative means heading back toward the origin.

DIFFERENTIABLE AT a

$$f \text{ is differentiable at } a \iff f'(a) \text{ exists}$$

For example: $|x^2 - 4|$ flips branches where $x^2 - 4$ is negative, i.e. $|x| < 2$ — so $f' = 2x$ outside and $-2x$ inside. Two corners, at $x = \pm 2$: differentiable on $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$

Kinds of non-differentiability

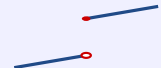
- ① **corner** f' **jumps** — the one-sided slopes are not equal
- ② **cusp** f' has a **vertical asymptote** — slopes get infinite
- ③ **jump in f** f' is **not there at all** — f is not even continuous
- ④ **vertical tangent** f' runs to $\pm\infty$ — the tangent is vertical, so it has no slope



corner



cusp



jump



vertical tangent

WATCH OUT

- ✗ $f'(2) = 8$ is the equation of the tangent line $\rightarrow$ it is the *slope*, not the TL
- ✗ the difference quotient is the derivative $\rightarrow$ that is the *secant* slope — take the limit
- ✗ continuous means differentiable $\rightarrow |x|$ is continuous at 0 and not differentiable there
- ✗ corner = cusp $\rightarrow$ corner: f' has a *jump discontinuity*. cusp: f' has a VA
- ✗ $\frac{dy}{dx}$ is one symbol $\rightarrow dx$ and dy are differentials